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B2308.C

```c
#include <stdio.h>
#include <conio.h>

#define MAX_SENSORS 5

struct SensorData {
    char zoneName[35];
    int moisture_level;
};

void selectionSort(struct SensorData sensors[], int n) {
    int i, j, min_idx;
    struct SensorData temp;

    for (i = 0; i < n - 1; i++) {
        min_idx = i;
        for (j = i + 1; j < n; j++) {
            if (sensors[j].moisture_level < sensors[min_idx].moisture_level) {
                min_idx = j;
            }
        }
```

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```
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        temp = sensors[min_idx];
        sensors[min_idx] = sensors[i];
        sensors[i] = temp;
    }
}

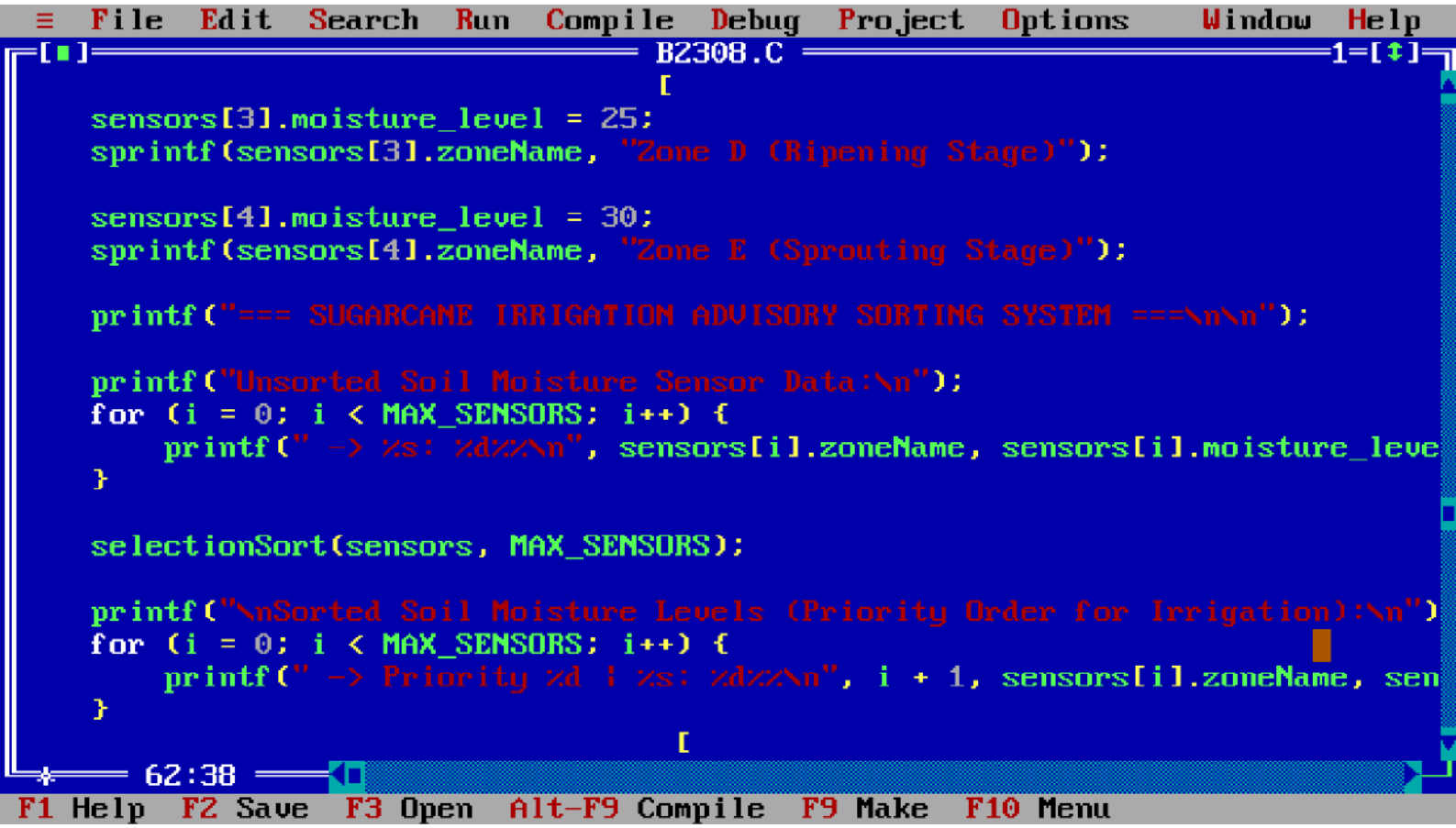
int main() {
    int i;
    struct SensorData sensors[MAX_SENSORS];

    clrscr();

    sensors[0].moisture_level = 40;
    sprintf(sensors[0].zoneName, "Zone A (Germination Stage)");

    sensors[1].moisture_level = 15;
    sprintf(sensors[1].zoneName, "Zone B (Tillering Stage)");

    sensors[2].moisture_level = 60;
    sprintf(sensors[2].zoneName, "Zone C (Grand Growth Stage)");
                                 [_
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```


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sensors[3].moisture_level = 25;
sprintf(sensors[3].zoneName, "Zone D (Ripening Stage)");
sensors[4].moisture_level = 30;
sprintf(sensors[4].zoneName, "Zone E (Sprouting Stage)");
printf("=== SUGARCANE IRRIGATION ADVISORY SORTING SYSTEM ===\n\n");
printf("Unsorted Soil Moisture Sensor Data:\n");
for (i = 0; i < MAX_SENSORS; i++) {
printf(" -> %s: %d%%\n", sensors[i].zoneName, sensors[i].moisture_leve
}
selectionSort(sensors, MAX_SENSORS);
printf("\nSorted Soil Moisture Levels (Priority Order for Irrigation):\n")
for (i = 0; i < MAX_SENSORS; i++) {
printf(" -> Priority %d : %s: %d%%\n", i + 1, sensors[i].zoneName, sen
}
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```
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    sensors[4].moisture_level = 30;
    sprintf(sensors[4].zoneName, "Zone E (Sprouting Stage)");

    printf("=== SUGARCANE IRRIGATION ADVISORY SORTING SYSTEM ===\n\n");

    printf("Unsorted Soil Moisture Sensor Data:\n");
    for (i = 0; i < MAX_SENSORS; i++) {
        printf(" -> %s: %d%%\n", sensors[i].zoneName, sensors[i].moisture_leve
    }

    selectionSort(sensors, MAX_SENSORS);

    printf("\nSorted Soil Moisture Levels (Priority Order for Irrigation):\n")
    for (i = 0; i < MAX_SENSORS; i++) {
        printf(" -> Priority %d | %s: %d%%\n", i + 1, sensors[i].zoneName, sen
    }
                                        [

    getch();
    return 0;
}                                       [_
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```

```
=== SUGARCANE IRRIGATION ADVISORY SORTING SYSTEM ===

Unsorted Soil Moisture Sensor Data:
 -> Zone A (Germination Stage): 40%
 -> Zone B (Tillering Stage): 15%
 -> Zone C (Grand Growth Stage): 60%
 -> Zone D (Ripening Stage): 25%
 -> Zone E (Sprouting Stage): 30%

Sorted Soil Moisture Levels (Priority Order for Irrigation):
 -> Priority 1 | Zone B (Tillering Stage): 15%
 -> Priority 2 | Zone D (Ripening Stage): 25%
 -> Priority 3 | Zone E (Sprouting Stage): 30%
 -> Priority 4 | Zone A (Germination Stage): 40%
 -> Priority 5 | Zone C (Grand Growth Stage): 60%
```